CS 4476 PS 3

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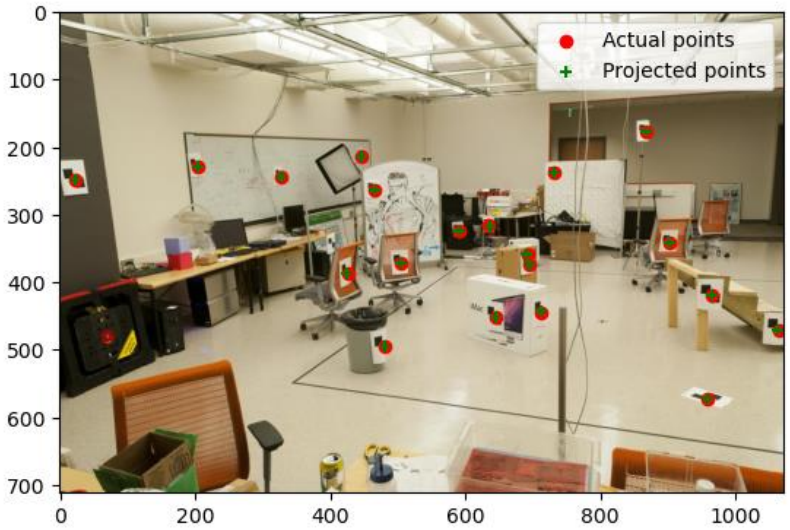
Part 1.5: Projection Matrix for provided image



3x4 projection matrix has 12 numbers in it, but its overall scale doesn't matter. This brings the number of actual unknowns down to 11. Each point match between the 3d world and the 2d image gives you two equations (one for x and one for y).

Therefore, to solve for 11 unknowns, you need at least 11 equations. Five points would only give you 10 equations, which isn't enough. The sixth point gets you to 12 equations, giving you enough information to lock down the solution for the matrix.

Part 2.1: Projection Matrix for custom images

Figure 1

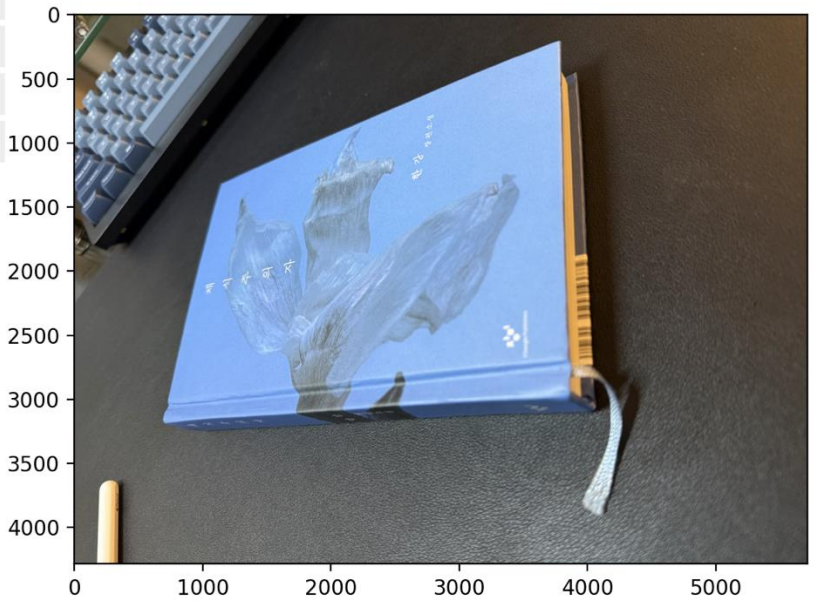
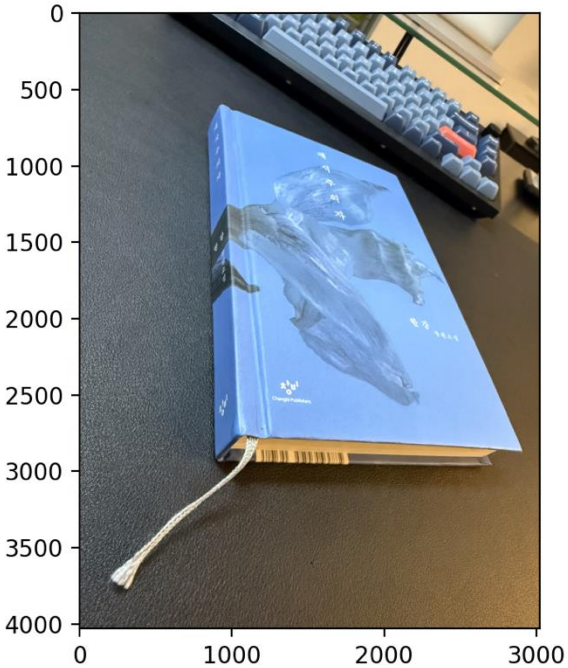
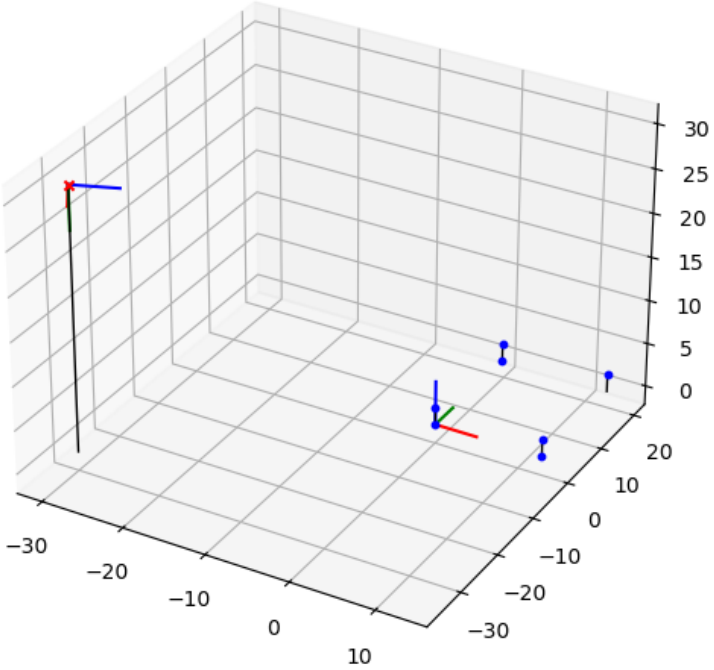


Figure 2

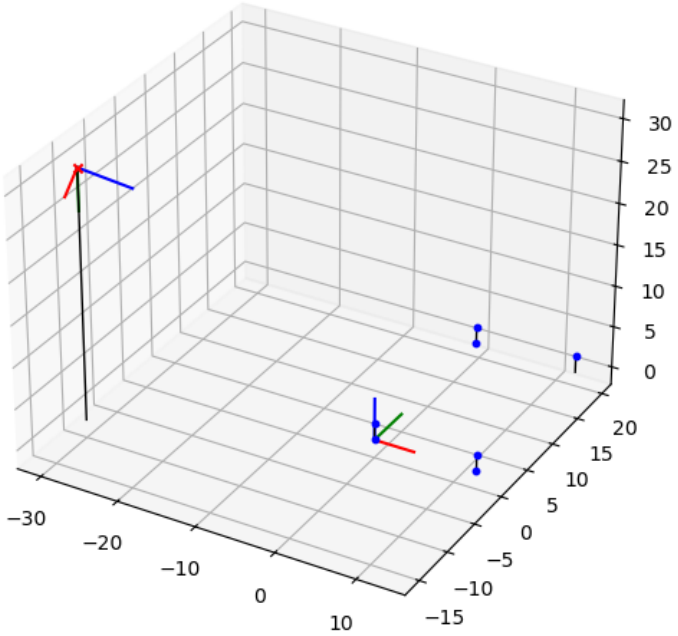


Part 2.2: Pose init for custom images

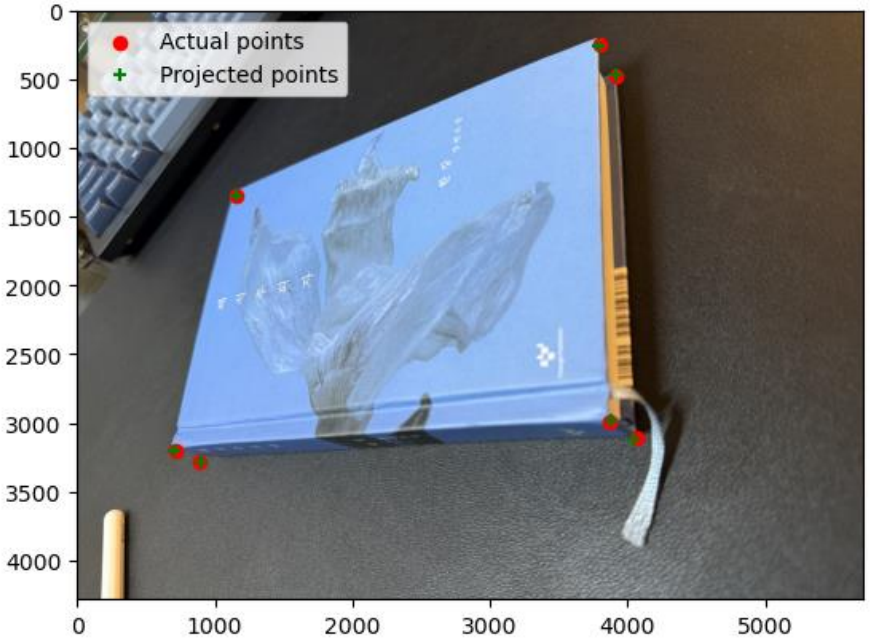
pose

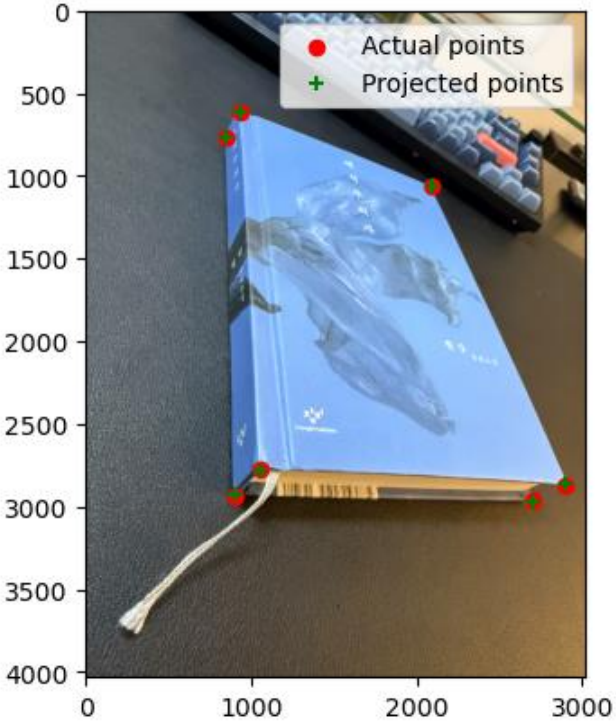


a pose for



Part 2.2: Optimized results for custom images





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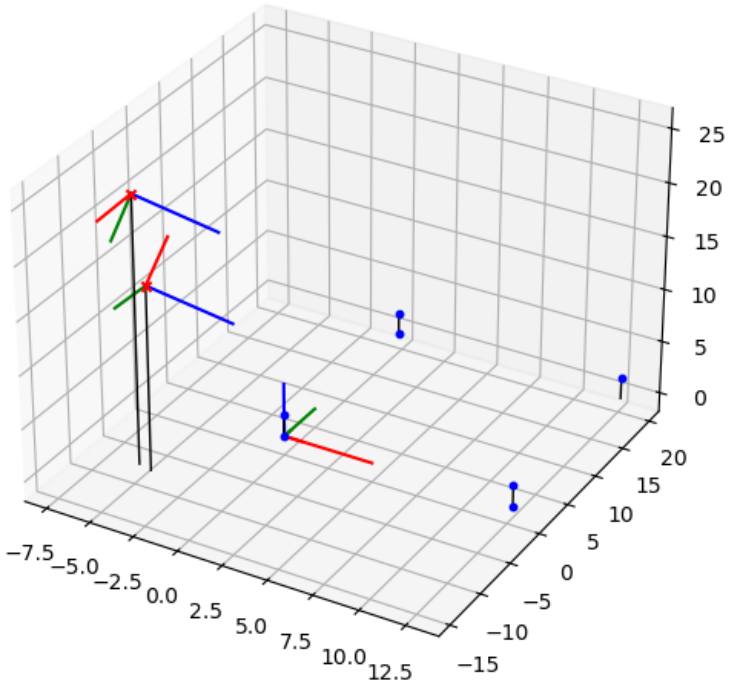
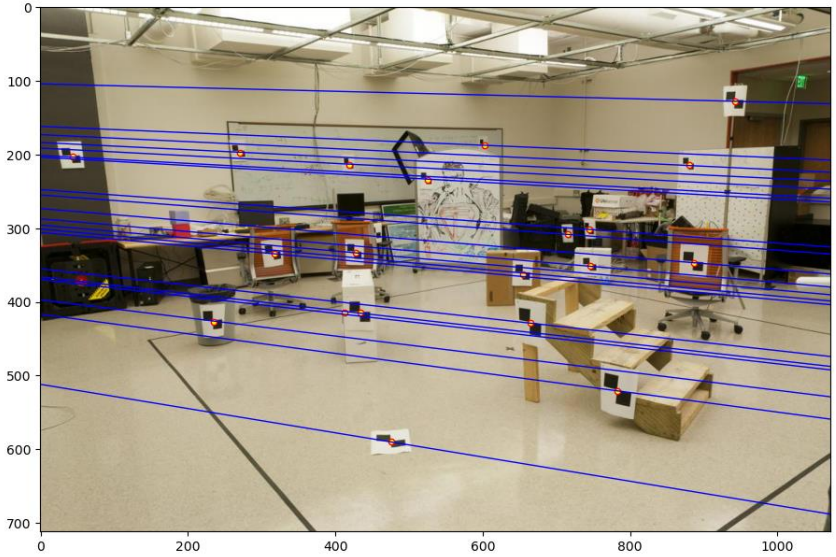
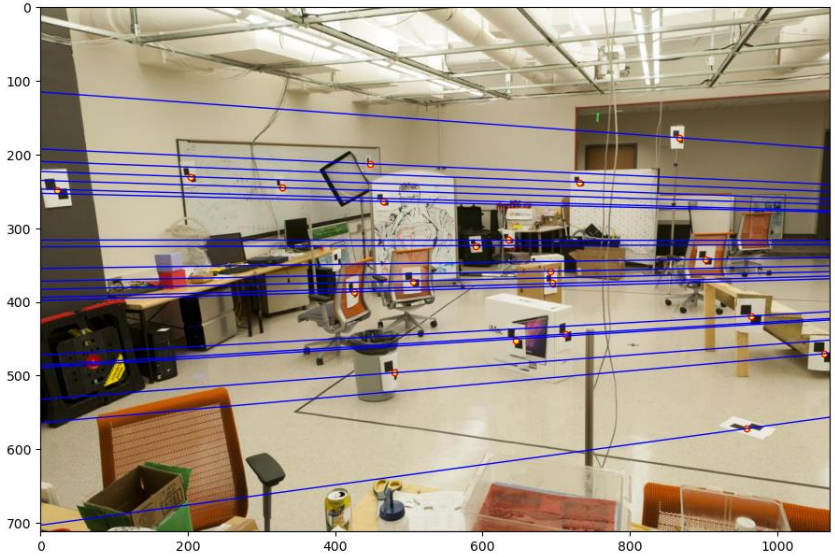
Part 2.3: Optimized Camera Poses



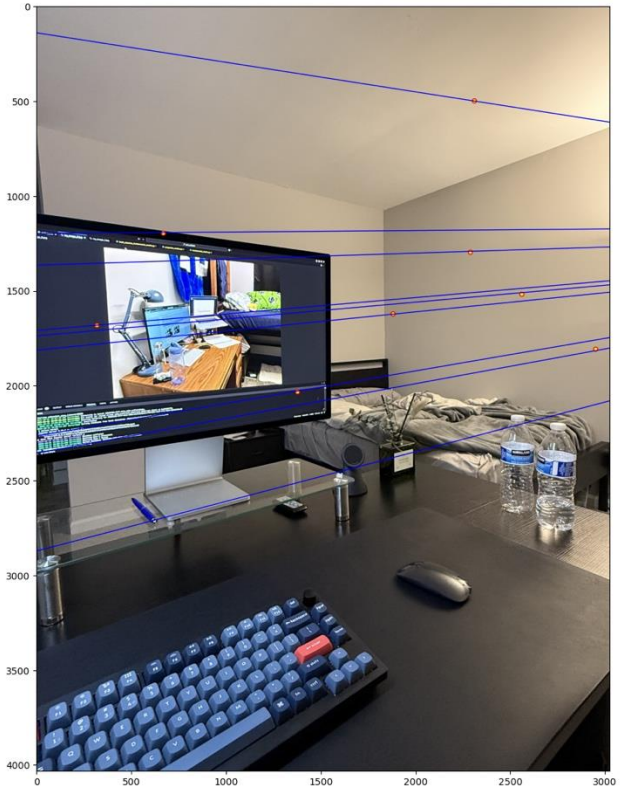
coordinate systems [1]>

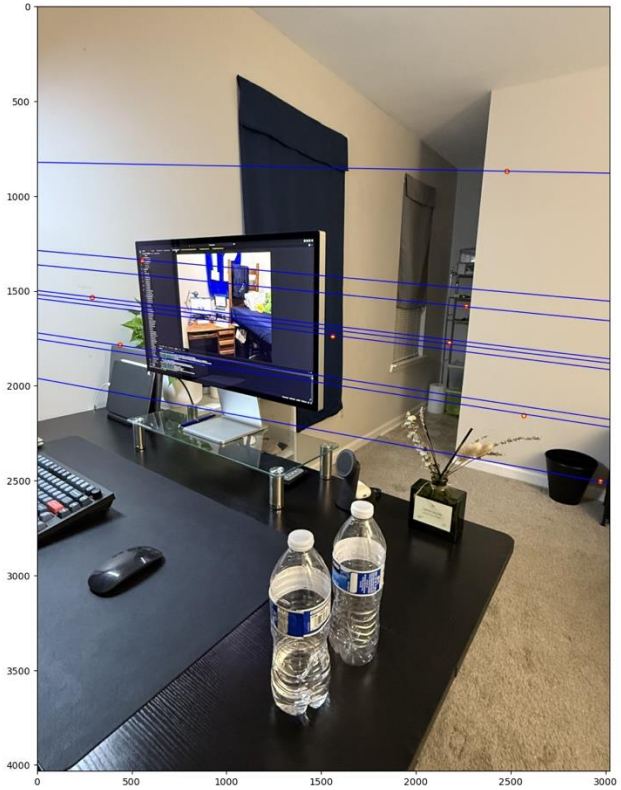
Part 3.2: Optimized Epipolar Lines (given images)





Part 3.3: Optimized Epipolar Lines (custom images)





epipolar lines> [1.5]

Part 3.4: Reflection Questions [1x3]

1. If you only rotate the camera, the two images are related by a homography, not by epipolar geometry. The camera's position doesn't change, so the baseline between the two views is zero. This makes the fundamental matrix ill-defined and prevents you from recovering any 3d structure or depth information from the scene.

2. A point in the first image corresponds to a ray in 3d space. The fundamental matrix projects this entire ray into the second image, creating the epipolar line. Without knowing the 3d point's depth (its exact location along that ray), you can't know exactly where on the epipolar line it will land. The projection from 3d to 2d loses this depth information.

3. If the camera centers are visible in each other's field of view, the epipoles will appear inside the image frames. Since all epipolar lines must pass through the epipole, the lines will radiate outwards from the epipole in each image, like spokes on a wheel.

Part 3.4: Reflection Questions [1x3]

4. This is a case known as stereo rectification. It means the two cameras are perfectly aligned: their image planes are parallel, and the baseline connecting them is horizontal. This setup is ideal for CV because it simplifies the search for matching points to a much faster 1d search along the same horizontal line in the other image.

5. The fundamental matrix is defined by the equation (x1.T)(Fx0)=0. If you multiply F by any non-zero number, the equation still holds true: (x1.T)(sF)x0 = s((x1.T)Fx0)=0. Since the core relationship doesn't change, the matrix is only defined up to a scale.

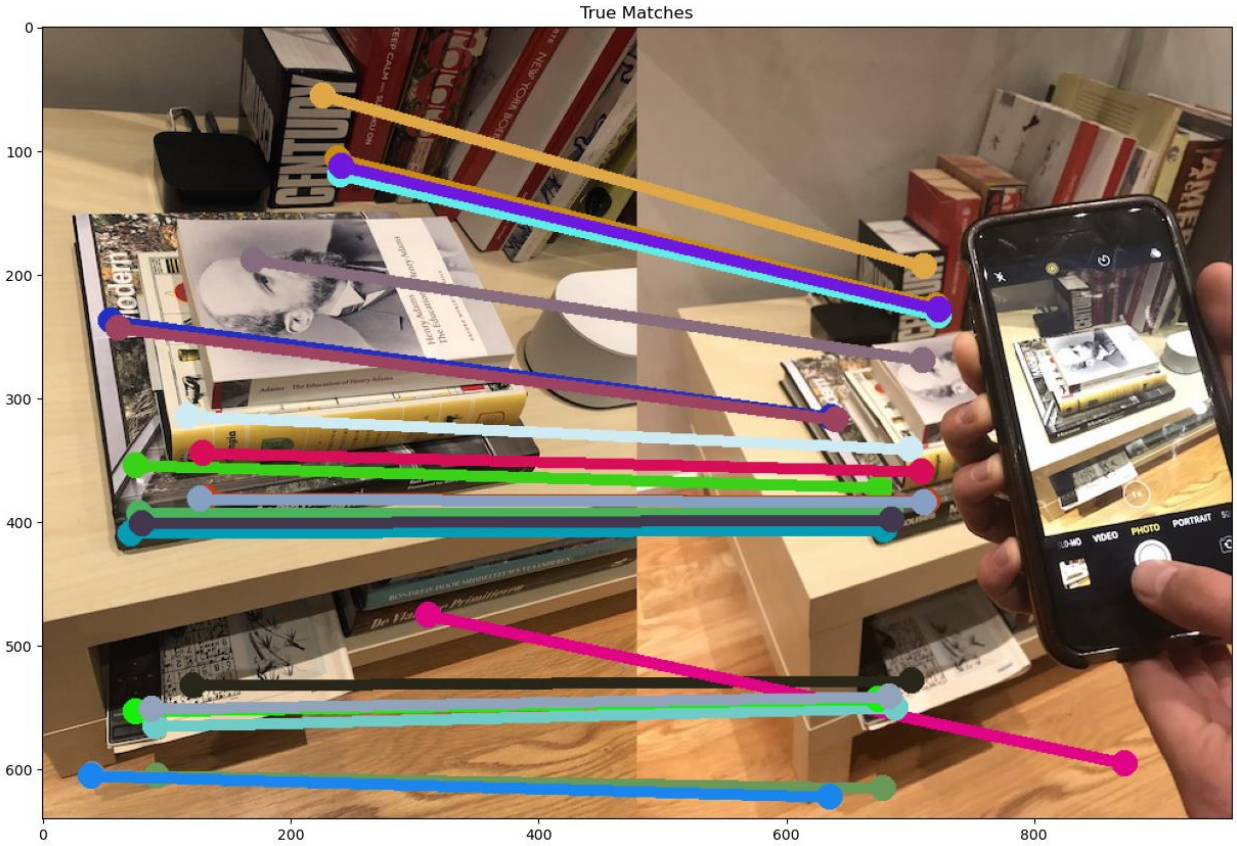
6. The rank 2 property is a mathematical consequence of epipolar geometry. Geometrically, it ensures that all epipolar lines in an image must intersect at a single point (the epipole). If the matrix were full rank (rank 3), the lines wouldn't be forced to converge, violating the geometric constraint that the epipole exists.

Part 4.2: RANSAC Iterations Questions [1x3]

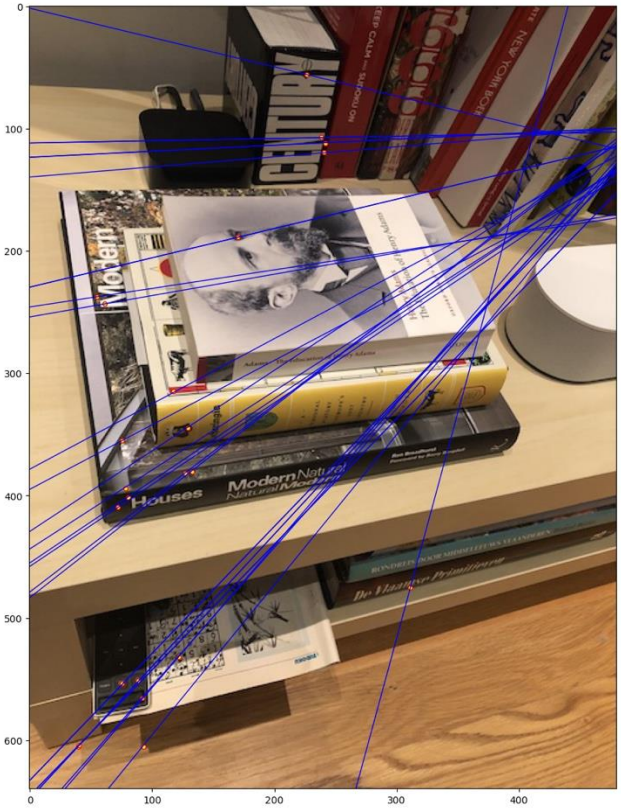
Type your answers to the three RANSAC Iterations questions from the jupyter notebook below:

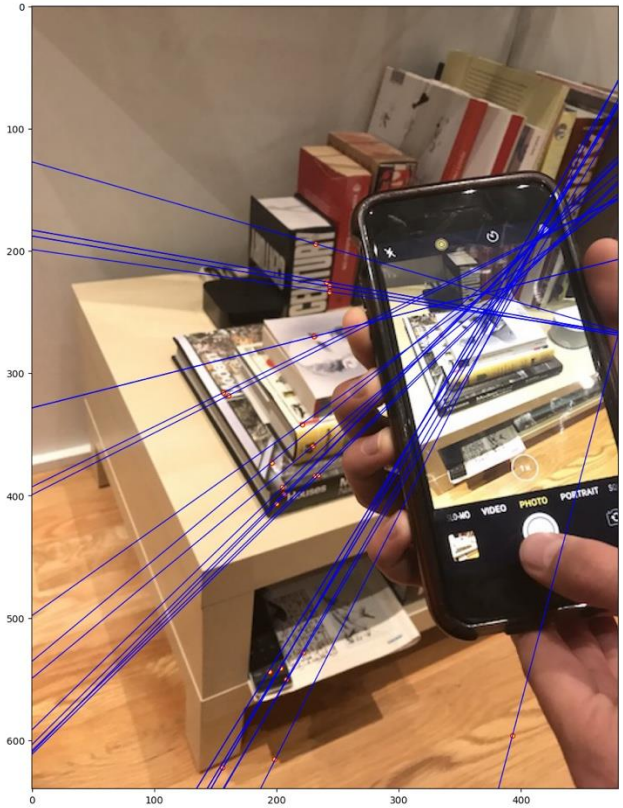
1. To find the fundamental matrix with 99.9% certainty from matches that are 90% accurate, using a sample size of 9 points, you would need a minimum of 15 RANSAC iterations: P (Success Probability): 0.999, p (Inlier Probability): 0.9, k (Sample Size): 9, s = log(1−P)/log(1−p**9) = 14.1
2. Using 18 points per sample would require a minimum of 43 RANSAC iterations. This demonstrates a key RANSAC principle: it's better to use the minimum number of points needed to fit the model. The probability of randomly selecting a sample with zero outliers decreases dramatically as the sample size grows, requiring far more iterations to achieve the same level of confidence.
3. If the dataset accuracy drops to 70%, you would need a minimum of 81 RANSAC iterations to find the fundamental matrix with 99.9% certainty, using a sample size of 7 points. P (Success Probability): 0.999, p (Inlier Probability): 0.70, k (Sample Size): 7, S = (1−0.999)/ log(1−0.70**7)log = 80.4

Part 4.4: RANSAC Inlier Matches



Part 4.4: RANSAC Epipolar Lines





nage with epipolar

Local Unit tests results

```
(ps3) lagn1@Evans-MacBook-Pro ps3_unit_tests % pytest ./.
========================================================== test session starts ==========================================================
platform darwin -- Python 3.9.24, pytest-8.4.2, pluggy-1.5.0
rootdir: /Users/lagn1/Desktop/ps3_release
plugins: anyio-4.10.0
collected 20 items

part1_unit_test.py .....                                                                                                          [ 25%]
test_essential_matrix_decomposition.py ..                                                                                         [ 35%]
test_fundamental_matrix.py .........                                                                                              [ 80%]
test_ransac.py ....                                                                                                               [100%]

========================================================== 20 passed in 0.70s ==========================================================
(ps3) lagn1@Evans-MacBook-Pro ps3_unit_tests %
```

Conclusions

<Describe what you have learned in this project. Feel free to include any challenges you ran into.> [2]

A major challenge I faced was with the unit tests, particularly test_ransac_fundamental_matrix_fit. My initial code was logically correct but failed because it wasn't numerically stable. I learned that tuning parameters like the inlier threshold is critical for getting an accurate final result, as the suggested value was too lenient for one of the tests.